

Does curbside composting increase how much food households throw away?

Abstract

Expanding municipal curbside composting is a common policy for diverting organic material from landfills, yet its impact on household food waste is uncertain. This study investigates whether curbside composting induces a behavioral “rebound effect,” leading households to waste more food due to reduced moral or financial costs. Utilizing cross-sectional data from the 2022 OECD Environmental Policy and Individual Change (EPIC) household survey across nine countries, we construct a food waste index based on self-reported discard frequencies. To establish causality and account for selection bias in composting service provision, we employ clustered propensity-score matching and Lewbel instrumental variable regressions. By regressing reported mixed waste volumes on our index, we translate behavioral shifts into physical quantities. Using our preferred estimator, we find that curbside composting services cause a modest increase in the volume of food households discard (either as compost or mixed waste), with country-level estimated volumetric effects ranging from a 0.2% (in Israel) to 2% (in Belgium) increase in food waste volumes. We also recover novel, country-level estimates of the proportion of residential mixed waste volume that food waste takes up for non-composting households. These results provide the first large-scale empirical evidence of a composting rebound effect. We discuss the implications of such effects for municipal composting programs.

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1 Introduction

Food waste has been a growing concern for policymakers in many countries (Schanes et al., 2018; Thyberg and Tonjes, 2016). For example, over a third of food in the United States is wasted, with a significant portion winding up in landfills (U.S. Environmental Protection Agency, 2025). One policy emphasized for addressing this challenge and its environmental impacts is the expansion of food scrap collection and composting services (Farhidi et al., 2022). In some cases, these services are publicly offered by municipalities, at times accompanied by mandates for households to separate food scraps from other mixed waste. In other instances, households may voluntarily opt in to fee-based composting services provided by commercial operators.

Significant uncertainties remain regarding the effectiveness of composting services in improving food waste management, including their effects on overall food waste generation, levels of separation and recovery, and their environmental benefits. In this study, we analyze how household food waste generation responds to the provision of composting services. Several behavioral responses are possible, including effects that are analogous to the ‘rebound effect’ studied in the energy efficiency literature (e.g., Chan and Gillingham, 2015): To the extent that households face ethical or moral costs from discarding food with the rest of their trash, composting offers a means to reduce these ‘moral costs’ of throwing away food. This type of response corresponds closely to a ‘behavioral rebound effect’ described and investigated by Dorner (2019), in which technological improvement induces a reduction in pro-environmental effort through a reduction in marginal environmental damages and ‘moral licensing.’ Additionally, when unit-based (‘pay-as-you-throw’, PAYT) mixed waste charging schemes are also present, the provision of food waste collection services decreases households’ financial costs of discarding food, regardless of any moral costs.

This study provides the first large-scale analysis of how households’ decisions to discard food are affected by public or private provision of food scrap separation services. We use novel data from a recent large-scale household survey administered in nine OECD countries (OECD, 2023). The survey included an extensive section on household waste generation and behavior and also elicited information regarding the structure and extent of available waste collection and separation services.

While these cross-sectional survey data provide a rich, novel source of information for addressing our research question, causally identifying the effect of food scrap collection on food waste generation is challenging for several reasons which we address in this work. First, significant selection effects may

be present, with food scrap collection services being systematically provided to households who are more or less predisposed to discard food. Fortunately, these data contain a rich set of household characteristics, including demographics, socioeconomic factors, and environmental attitudes, which we use to control for selection on observables. We do so using conventional linear regression analysis and matching estimators. To retain statistical power while acknowledging potentially significant and confounding between-country heterogeneity, we include country fixed effects in the regression analysis. In matching estimation, we employ a clustered propensity-score matching (CPSM) method, which combines PSM with exact (or near-exact) matching on the cluster variable, in this case, country (Arpino and Cannas, 2016). To our knowledge, this is a novel application of this method in environmental economics. Additionally, to test for potential endogeneity (*e.g.*, due to selection on unobservables), we employ an instrumental variable (IV) method introduced by Lewbel (2012), wherein IVs are constructed using estimated heteroskedasticity in the selection equation.

A second challenge is that the survey did not measure food waste generation in terms of a physical quantity, but rather asked a series of questions about which types of foods households usually discarded. This potentially confounds the policy-relevance of the estimated causal effects, since knowing whether composting services affect households' choices about which foods they discard does not directly reveal effects on overall food waste volumes (the policy-relevant outcome). To address this, we first construct an aggregate food waste index from the stated food waste behavioral indicators; this index is the main outcome we analyze in causal inference. To tie this outcome to physical quantities, we then employ mixed waste volume measurements collected in the survey. By regressing mixed waste volume on this food waste index in the subsample of households without composting, we obtain a marginal effect of the food waste index on mixed waste volumes. We then use these volumetric marginal effects to convert our estimated effects of composting on food waste behaviors into physical quantities. This approach has the added benefit of generating estimates of the proportion of mixed waste occupied by food waste (at least for non-composting households).

We find that curbside composting service is associated with a modest increase in household food waste, on average equivalent to approximately three quarters of a liter per week. These findings provide the first large-scale empirical evidence that composting services increase household food waste generation. This evidence is important not so much as a critique of policies promoting curbside composting and food scrap collection, but rather to help planners anticipate behavioral

responses to these policies and ensuing changes to waste streams. Predicting such behavioral responses can help planners more accurately gauge the volumes of compost and mixed waste that municipalities will need to handle following implementation of curbside programs.

The rest of this paper proceeds as follows. First, we review existing literature on the links between food waste generation and composting, which we use to form our central hypothesis tested in this study. We describe the data we use to test our hypothesis and the econometric methods we use to mitigate potential threats to causal inference, including clustered propensity score matching and Lewbel IV methods. We also detail how we infer volumetric effects from survey indicators of household food waste behavior. We then share the estimation results and discuss their implications for future research and policy.

2 Related literature and tested hypothesis

Despite the clear importance of economic and behavioral feedback mechanisms in understanding the effects of composting policies, there is very limited prior quantitative research assessing the effects of composting service provision on food waste generation. Hegwood et al. (2023) use computable general equilibrium models to analyze the potential effects of food system efficiency improvements that reduce food waste, and conclude that classical food consumption rebound effects associated with such efficiency gains could offset 53-71% of avoided food waste. To our knowledge, the only published empirical study assessing the effects of composting on food waste generation is provided by Qi and Roe (2017). They conducted a small experiment with composting in a university cafeteria and found that cafeteria diners threw away more food when they were informed that their food waste would be composted. A correlational study by van der Werf et al. (2020) also found participation in curbside organic waste collection in Canada was associated with a higher level of household total food waste. While Somers and Li (2025) find that access to curbside composting increased food purchases in households, they do not explicitly investigate curbside composting’s impact on food waste. By contrast, De Silva and Taylor (2024) find that the implementation of curbside composting in Australian municipalities had no appreciable effect on total waste. Thus, the effects of composting on food waste in quasi-experimental contexts are mixed, but the evidence is sparse.

Researchers have also been interested in how the disposal method impacts the amount of waste generated in the recycling domain. It is unclear how much consumers consider material reuse and waste reduction as complementary and substitute efforts. In a survey-based study, researchers

find that recycling and waste reduction behaviors tend to strengthen each other (D’Amato et al., 2016). These survey results are supported by data analysis on recycling and waste generation in Italy; Degli Antoni and Vittucci Marzetti (2019) find a 10% increase in the recycling rate of a locality reduces solid urban waste per capita by about 2%, and, more specifically, curbside recycling collection results in about a 4 percentage point decrease in waste generation per capita. Other studies have found the opposite. For example, Maier et al. (2023) finds that the availability of curbside recycling in North Carolina municipalities increases the total amount of waste generated, but staggered adoption difference-in-difference methods suggest imprecise estimates.

While rebound effects from energy efficiency improvements can be understood as a natural demand response to lower effective marginal costs of energy brought about by efficiency improvements, rebound effects in residential waste generation from recycling or composting likely extend beyond monetary incentives. This is because most households are not subject to unit-based (e.g. volumetric) charges for their mixed waste collection, and thus face a zero marginal financial cost for their waste generation.

Instead, several behavioral mechanisms may explain waste rebound effects and other types of behavioral rebound effects. In a laboratory experiment involving contributions to a tree-planting charity, Dorner (2019) offers two possible explanations for the behavioral rebound effect he studies: his first explanation as applied to our context is that, to the extent that households weigh the environmental damage externalities in their decisions to discard food, composting can be conceived as reducing the marginal environmental damages of their decision by diverting discarded food from the landfill. The second channel Dorner (2019) theorizes for behavioral rebound effects is moral licensing, which reduces feelings of guilt associated with an antisocial behavior (here, throwing away food) through engaging in a prosocial behavior (here, composting). Other behavioral explanations have been offered for rebound effects observed in household recycling, including ‘warm-glow’ effects (van Doorn and Kurz, 2021).

In this study, we are not able to isolate which of these mechanisms (or others) may explain food waste rebound effects in response to composting. We simply offer these multiple explanations as comprising a clear theoretical motivation for the hypothesis we aim to test, which is the wide-spread existence of such an effect.

3 Data

The data we use to test our hypothesis come from the 2022 iteration of the OECD’s Environmental Policy and Individual Change (EPIC) surveys (OECD, 2023); details about this data source are provided in the supplementary material. These surveys are implemented as web-panels using quota sampling based on age, gender, household income, and subnational region, and are designed to be representative at the country-level. The 2022 survey round included nine OECD countries (USA, United Kingdom, France, Belgium, Sweden, Israel, Switzerland, the Netherlands, and Canada). The full survey dataset consists of 17,670 unique household responses, with a subset of 8,569 households randomly selected from the full sample to answer a section containing the waste generation and recycling questions. The survey elicited the availability of different waste collection and recycling services, including the availability of compost collection services and total household waste generation volume. While the survey did not elicit aggregate household food waste volumes, it asked respondents whether they threw away several types of food, as well as the number of bags (of defined volume) of mixed waste disposed of weekly.

Table 1 summarizes by country, the proportion of respondents with no composting and curbside composting. When we exclude drop-off composting services (not shown in the table), the sample size drops to 5,838. The availability of curbside composting service varies significantly across countries; only 9% of Israeli respondents, but 63% of Belgian respondents have access to curbside composting.

We construct a food waste index based on the indicators of which food types were discarded. Survey respondents were asked: “What type of food does your household usually throw away? Please exclude non-edible parts of food, *e.g.* peelings, apples cores?” The six categories of food included in this question were: fruits/vegetables, bread, meat/seafood, dairy/eggs, packaged food, and leftovers. We define a respondent’s food waste index as the proportion of the categories that they said they usually threw away; for example, if a respondent indicated that they usually threw away bread and dairy/eggs, then their food waste index is $2/6$, or 0.3333.

We also use an extensive array of attitudinal variables collected in the survey, specifically a series of survey questions related to environmental attitudes. Responses to these environmental attitudinal questions provide a valuable set of covariates with which we can further control for selection into curbside composting. To reduce collinearity in these covariates, we use a latent class analysis (LCA) previously conducted by the OECD (Brown, 2024) to cluster respondents into groups according

to their stated environmental attitudes (Hagenaars and McCutcheon, 2002). The survey questions used in the LCA are:

1. “How important is climate change (e.g. rising average temperatures, extreme weather events or other environmental issues (e.g. pollution)) to you personally?”, with responses on 5-point Likert scale from “Not at all important” to “Very important”
2. Elicited agreement with eight different statements about environmental problems (Table 2): “To what extent do you agree with each of the following statements?” with responses on a 5-point Likert scale from “Strongly disagree” to “Strongly agree”.

The LCA by Brown (2024) estimated the probability that survey respondents belonged to four, mutually exclusive attitudinal classes, labeled as follows: *environmental prioritizers*, *technological optimists*, *economic prioritizers*, and *environmentally neutral*. See the supplementary material for further discussion of the characteristics of these classes in relation to the attitudinal questions.

In addition to varying composting service levels, there is also significant variation in mean food waste scores, ranging from 0.1359 to 0.4017 for households with no composting to 0.1861 to 0.4347 for households with curbside composting. For all nine countries, the mean food waste score is higher for curbside composting versus no composting, although it only reaches statistical significance for the Netherlands, France, the UK, and the USA. The key variables for our analysis are the food waste index discussed in the methodology section, composting service availability, demographic variables, and the environmental attitudes LCA classifications described above.

4 Methods

The econometric objective of this paper is to evaluate the causal effect of curbside composting services on household food waste generation. Because we only have data from a cross-sectional survey, we are limited in the causal inference methods we can use to answer this question. We begin with an ordinary-least squares (OLS) regression containing a rich array of covariates and country fixed effects, before turning to matching and instrumental variable (IV) methods.

We first estimate an ordinary least squares (OLS) regression equation of the following form:

$$F_{i,c} = \tau C_{i,c} + \beta_W \mathbf{W}_{i,c} + \beta_X \mathbf{X}_{i,c} + \beta_A \mathbf{A}_{i,c} + \alpha_c + \epsilon_{i,c} \quad (1)$$

where $F_{i,c}$ is the food waste index value for household i in country c , $C_{i,c}$ is a binary variable

indicating whether the household has a curbside composting service, $\mathbf{W}$ is a vector of other characteristics about the local waste management system (including recycling service availability and volumetric mixed waste charges), $\mathbf{X}$ is a vector of household socioeconomic characteristics, $\mathbf{A}$ is a vector of variables measuring households' environmental attitudes, and α_c is a country fixed effect. The regression coefficient of interest is τ , i.e., the average treatment effect (ATE) of curbside composting service $C_{i,c}$ on food waste $F_{i,c}$. Because the data used for this model are cross-sectional, we cannot use panel regression methods (*e.g.*, household-level fixed effects).

4.1 Matching

We also use matching methods as an alternative way to account for treatment selection based on observable factors. Given the relatively small sample size, we primarily use propensity score matching (PSM) methods, rather than nearest-neighbor (NN) matching with a Mahalanobis distance metric or exact matching on binned data. Due to the cross-national structure of the survey, we use a clustered PSM (CPSM) technique from Arpino and Cannas (2016). CPSM seeks untreated matches for treated observations that come from the same cluster (in this case, the same country). This has the advantage of controlling for additional unobserved, cluster-level characteristics that may influence outcomes; for example, cultural norms regarding waste may be correlated with country of residence. Arpino and Cannas introduce two versions of clustered matching: (1) strict matching only from within the same cluster and (2) a 'preferred-within' procedure in which matches from the same cluster are given priority, before drawing matches from outside clusters when no feasible within-cluster matches are found. In a simulation exercise, Arpino and Cannas show that both 'within-cluster' and 'preferred-within' PSM perform better than simply including cluster fixed effects in the propensity score model.

The formula for the average treatment effect on the treated (ATT) using PSM is given as follows:

$$\tau^{ATT} = \frac{1}{|M|} \sum_{ic \in I_1 \cap M} \left(F_{ic} - \sum_{jc' \in I_0} F_{jc'} w(ic, jc') \right)$$

where I_0 and I_1 are the sets of control and treated individuals, respectively, M is the set of treated individuals with feasible matches (discussed momentarily), and $w(ic, jc')$ is the weight for the potential outcome $F_{jc'}$ of control individual j in country c' . When jc' is not matched to ic , then weight $w(ic, jc')$ is zero. Otherwise, the weight will be the reciprocal of the number of matches for ic . Matching each treated individual to multiple control individuals may decrease the estimator

variance; for many of the matching models assessed in this paper, treated individuals are matched to the $m = 2$ closest control individuals. Matching with replacement allows control individuals to be matches for multiple treated individuals.

Using notation derived from Arpino and Cannas (2016) and Abadie and Imbens (2016), we define the set of control group matches for ic as follows:

$$I_M(ic) = \left\{ jc' \in I_0 : \left(\sum_{k \in I_0} 1_{[|p(X_{ic}) - p(X_{kc'})| \leq |p(X_{ic}) - p(X_{jc'})| < \text{caliper} \cdot \hat{\sigma}_p]} \right) \leq m \right\} \quad (2)$$

where $p(X_{ic})$ is the propensity score estimate for individual i based on the covariates X . In words, we find the control individuals with the m closest propensity scores as long as the distance between the propensity score is less than some maximum distance which we set to be the caliper times the standard deviation of the estimated propensity score $\hat{\sigma}_p$. Then the matched set M is:

$$M = \{ic : I_M(ic) \neq \emptyset\} \cup \{\cup_{ic} I_M(ic)\} \quad (3)$$

So far, the equations describe the standard pooled matching approach. Clustered matching within country differs from the standard PSM by additionally requiring that the set of control group matches for ic comes from the same country cluster, i.e., $c = c'$:

$$I_M^{Within}(ic) = \left\{ jc' \in I_0 : \left(\sum_{k \in I_0} 1_{[|p(X_{ic}) - p(X_{kc'})| \leq |p(X_{ic}) - p(X_{jc'})| < \text{caliper} \cdot \hat{\sigma}_p]} \right) \leq m; c = c' \right\} \quad (4)$$

Unfortunately, strict within-country matching can lead to significant numbers of unmatched treated individuals being dropped if there is limited common support between control and treated individuals within each country. Arpino and Cannas (2016) offer a compromise between pooled and strict within-cluster PSM: ‘preferred within-cluster’ matching, which requires searching for matches that meet caliper requirements within the same country cluster c ; only if suitable matches cannot be located in the same country cluster c , then we attempt to identify matching control individuals (within caliper tolerances) in other clusters.

4.2 Lewbel instrumental variables regression

As a further validity test, we use instrumental variable (IV) regressions. Since no suitable IV for curbside composting service could be identified directly in the available survey data, we employ a technique introduced by Lewbel (2012). This technique uses estimated heteroskedasticity from a regression of the potentially endogenous variable on the exogenous variables to create instruments.

Following Lewbel’s model, we have two equations. The first equation has the same structure as the OLS regression eq. (1), containing the effect of interest τ (the impact of curbside composting C on food waste F) and controlling for the same observables (including country dummies $\mathbf{d}$) $\mathcal{X} := [\mathbf{W}, \mathbf{X}, \mathbf{A}, \mathbf{d}]$:

$$F_{ic} = \tau C_{ic} + \beta \mathcal{X}_{ic} + \epsilon_{1ic} \quad (5)$$

The second is a selection equation, regressing curbside composting C on the same controls as in the first equation:

$$C_{ic} = \alpha \mathcal{X}_{ic} + \epsilon_{2ic} \quad (6)$$

Lewbel shows that under certain assumptions, the variables $(\mathbf{Z} - \bar{\mathbf{Z}})\hat{\epsilon}_2$ are valid IVs for C , where $\mathbf{Z} \subseteq \mathcal{X}$: These instruments satisfy the exclusion restrictions when $\mathbb{E}[\mathcal{X}\epsilon_1] = \mathbb{E}[\mathcal{X}\epsilon_2] = \text{cov}(\mathbf{Z}, \epsilon_1\epsilon_2) = \mathbf{0}$. The instruments are powerful when the covariate subset $\mathbf{Z}$ is heteroskedastic in the selection equation, *i.e.*, $\text{cov}(\epsilon_2^2, \mathbf{z}) \neq 0$ for all $\mathbf{z} \in \mathbf{Z}$. We thus select $\mathbf{Z}$ by, first estimating eq. (6) via OLS and then taking the squared fitted residuals $\hat{\epsilon}_2^2$ from that regression and regressing them on the controls $\mathcal{X}$; each covariate with an estimated coefficient significant at the 0.05-level is included in $\mathbf{Z}$. We then use these Lewbel IVs in an otherwise-standard two-stage least squares (2SLS) estimation of eq. (5).

4.3 Volumetric effects

An additional data limitation we address is that, as described above, our measure of food waste from the survey is an index constructed from a set of stated behaviors, whereas impact evaluation in terms of food waste weight or volume is more directly relevant for informing waste management policy. To obtain estimated effects in volumetric terms, we use additional data reported in the survey on the volume of mixed waste generated by the respondent’s household, V . We regress the log of these volumes on the food waste index $F_{i,c}$ (and controls), for the subsample of observations lacking composting service (of any kind), based on the assumption that all food waste in such households must then be accounted for in mixed waste.

This regression takes the following general form:

$$\log V_{ic} = \delta_c f(F_{ic}) + \gamma \mathbf{X}_{ic}^V + \kappa_c + \nu_{ic} \quad (7)$$

where F_{ic} is the food waste index, the function $f(\cdot)$ is a potentially nonlinear transformation of this index, $\mathbf{X}_{ic}^V$ is a set of control variables in this regression, κ_c is a country fixed effect, and ν_{ic} is

regression error. The country-specific slopes δ_c on the transformed food waste index $f(F_{ic})$ allow the relationship between this index and mixed waste volumes to vary by country. We evaluate two alternative transformations of the index: the identity function $f(F) = F$ and the concave function $f_\alpha(F) = \log(1 + \alpha F)/\alpha$, where $\alpha(> 0)$ determines the degree of nonlinearity in the transformation. Note $f_\alpha(\cdot)$ becomes linear in the limit ($f_\alpha(F) \rightarrow F$) as $\alpha \rightarrow 0$. When using the transformation function $f_\alpha(\cdot)$, we estimate α along with the other parameters using nonlinear least squares (implemented via GMM).

Using the volumetric regression equation above, we can estimate two useful quantities related to our research question. First, as a ‘sanity check’, we are able to infer the percentage of food waste in each household’s total mixed waste volume (among non-composting households). Non-food mixed waste volume V_{ic}^0 can be calculated from the above regression equation by setting $F_{ic} = 0$ (i.e., indicating the household discards no food), yielding:

$$V_{ic}^0 = \exp(\gamma \mathbf{X}_{ic}^V + \kappa_c + \nu_{ic}) \quad (8)$$

The volume of food waste V_{ic}^F is then given by the difference between the observed and non-food waste volumes, i.e., $V_{ic}^F = V_{ic} - V_{ic}^0$. Food waste’s percentage ρ_{ic}^F of total mixed waste volume can thus be computed as follows:

$$\rho_{ic}^F = (V_{ic} - V_{ic}^0)/V_{ic} = 1 - \exp(-\delta_c f(F_{ic})) \quad (9)$$

Second, and most relevant to our research aim, given an ATT estimate of curbside composting’s effect on the food waste index, we are able to infer the volumetric effects as follows. Extrapolating the volumetric regression above to the full sample, the volume of food waste among the treated can be inferred as before using the difference between the observed and counterfactual no-food-waste volumes: $V_{ic}^F = V_{ic} - V_{ic}^0$. Among treated units, the counterfactual level of the food waste index (i.e., in the absence of curbside composting) is $\tilde{F}_{ic} = F_{ic} - ATT$. The resulting counterfactual food waste volume is therefore:

$$\tilde{V}_{ic}^F = \exp(\delta_c f(\tilde{F}_{ic})) + \gamma \mathbf{X}_{ic}^V + \kappa_c + \nu_{ic} \quad (10)$$

The estimated percentage change in food waste volume $\Delta\rho_F$ in response to curbside composting among treated units can therefore be inferred as:

$$\Delta\rho_{ic}^F = (V_{ic}^F - \tilde{V}_{ic}^F)/V_{ic}^F = \exp(\delta_c [f(F_{ic}) - f(\tilde{F}_{ic})]) - 1 \quad (11)$$

Below, we compute and report country-level means of this quantity.

5 Results

We first present and discuss the estimated effect of curbside composting on household food waste behaviors. We then present and discuss inference regarding volumetric effects.

5.1 Curbside composting effects on the food waste indicators

The estimated treatment effect of curbside composting on the food waste index of the treated group across different matching models is shown in Table 3. There are three propensity score models which differ by how matches are clustered within country. The model in column 1 incorporates country as a fixed effect in the propensity score model, but otherwise does not explicitly match treated individuals with control individuals in the same country. The model in column 2 does not include country in the propensity score model but requires matches to be within the same country; matches could not be found within country for a relatively large number of observations resulting in 458 observations being dropped. The model in column 3 also matches within countries but retains more observations by allowing matches outside of the country if a suitable within-country match cannot be found. Column 4 shows the results for a nearest-neighbor (NN) model using a Mahalanobis distance metric; even though only a subset of the propensity score matching variables was used in the nearest neighbor matching, over a quarter of the sample was dropped because it was not possible to find suitable matches across the matching variables. The estimates for τ_t have a similar magnitude across the three propensity score models, between 0.02 and 0.04; the NN model estimate for τ_t is smaller and is not statistically different from zero. The matching estimates are similar to what is estimated using an ordinary least squares (OLS) regression with country fixed effects (col. 5 in Table 3).

When we look at the underlying logistic regression results used to compute propensity scores (see Table S1 in the supplementary material), we see evidence for selection on observables. Curbside composting appears significantly associated with: the availability of ‘pay-as-you-throw’ mixed waste pricing, middle-aged respondents, higher household income, larger households and those with children, homeowners (v. renters), living in small towns, as well as those respondents not classified as “economic prioritizers” based on their stated environmental attitudes. These estimated associations suggest that households’ access to curbside composting is nonrandom, associated both with locational factors and local waste management policies and with individual household characteristics and attitudes. While we lack more detailed information about the variability in curbside

programs between and within countries, these associations are consistent with either mandatory curbside composting in some instances (based on significant associations with local governmental policies and population density) and, in other instances, with the availability of voluntary composting programs (based on significant associations with age, income, household size, and individual environmental attitudes).

Our favored treatment effect estimate corresponds to the preferred within-country PSM model in col. 3 of Table 3. We prefer this model because it balances several desirable qualities: ensuring matches come from the same country when possible and that matches are not too distant from one another in their estimated propensity for treatment, while not dropping too many observations to maintain statistical power. This model also achieves good balance on the matching covariates, as shown in Figure 1. This figure plots mean standardized differences for the matching variables between the treatment and control groups before and after the matching procedure. The variables in the figure are ordered top-to-bottom in terms of the difference between treatment and control prior to matching. Starred variables indicate the use of means standardized by the covariate’s standard deviation in the treated group; the unstarred, categorical variables indicate the use of raw means. A cutoff of 0.1 for the standardized mean difference is a guideline for assessing balance and implying causal estimates. Post-matching, all covariates meet this criterion, with the exception of log household size.

Applying the Lewbel IV procedure (Section 4.2) as an alternative way to estimate the effect of curbside composting, we find that the control variables *renter*, *location* (*rural*, *suburban*, *urban*), and the country FEs satisfy the heteroskedasticity assumption in the selection equation (6) needed for instrument power. Performing 2SLS with these IVs, we find that the coefficient on curbside composting (col. 6 of Table 3) remains similar in magnitude to the other estimates from OLS and matching. Diagnostic tests on the 2SLS regression show that it performs well: The first stage F-statistic is 31.9, well above the rule-of-thumb first-stage F-stat > 10 needed for the IVs to be considered powerful. Because we obtain multiple IVs for curbside composting, we also perform overidentification tests of IV validity. The Sargan overidentification test statistic is 9.5320 (p-value=0.30): Technically, this means we do not reject the null hypothesis that, assuming one of the IV exclusion restrictions is valid, then they all are valid. The Wu-Hausman test statistic is 0.0686 (p-value=0.79), implying that we cannot reject the null hypothesis that curbside composting is in fact exogenous in eq. (5). This suggests that the 2SLS model unnecessarily increases the imprecision of

the ATT estimate (as noted by the ATT’s larger standard error in the 2SLS estimate, compared to OLS), without a gain in accuracy (*i.e.*, in econometric terms, if both OLS and 2SLS are consistent, then OLS is more efficient).

Both as a robustness check, and to further investigate the mechanisms behind composting’s impact on the food waste, we separately estimate effects on each of the six food waste categories included in the survey. Results are shown in the supplementary material in Table S4 and further discussed in section B.1; the estimates in this table are interpretable as treatment effects on the probability of the respondent’s household usually throwing away each type of food. Across nearly all econometric models we estimated (the only exception being for nearest-neighbor matching), these treatment effects show that curbside composting has the highest effect on households’ propensity to throw away leftovers, followed by fruit and vegetables, and then bread. This ordering of food-type-specific effects is intuitive, given that these are also the categories with the highest shares of households in the overall sample indicating they usually throw away these items (43% for leftovers, 33% for produce, and 26% for bread). These categories also make up the bulk of household food waste by weight (Parizeau et al., 2021; ReFED, 2025; Roe, 2026; WRAP, 2025).

While exhibiting an intuitive pattern, the heterogeneity in estimated effects on the different food type components in our food waste index raises a potential concern that the constructed index does not properly account for how each of these food types differentially contributes to overall food waste volume. To address this aspect, we modify our food waste index, by calculating a weighted sum of each component food type indicator. To calculate the weights, we use each food type’s previously estimated contribution to overall food waste volume, based on prior literature. We specifically base these weights on reported weight-based compositional food waste analysis by WRAP (2025) for UK households. Section B.2 in the supplementary material discusses the construction of this weighted index in more detail, and supplementary Table S5 shows the estimated ATTs using the weighted index. The results closely mirror those shown in Table 3 for the unweighted index.

5.2 Estimated volumetric effects

Summary results from the volumetric regressions are shown in Table 4, along with the inferred volumetric effects of curbside composting on food waste. Results are shown for two regression specifications, corresponding to different transformations of the food waste index F in the regressions (see Section 4.3). The best performing specification, according the Bayesian information criterion

(BIC) statistic, corresponds to the nonlinear transformation $f_\alpha(F) = \ln(1+\alpha F)/\alpha$. The underlying regression estimates can be found in Table S3 of the supplementary material.

Results from the best performing specification are shown in the last two columns of the Table 4. Combining the volumetric regression estimates with our curbside estimated index $ATT=0.04087$ (col. 3 of Table 3) implies that the volume of food discarded by households increased by between 0.2% (in Israel) to 2% (in Belgium). Note that the between-country variation in these volumetric effects reported in Table 4 derives both from country-specific slopes on the food waste index term estimated in the volumetric regression and from variation in the country-level differences in the baseline food waste index values for non-composting households (see eq. 11).

As a ‘sanity check’ on these volumetric estimates, we also report in Table 4 the implied percentage of food waste in overall household mixed waste volume, as per eq. (9). These estimates—which apply only to non-composting households—imply that food waste makes up 27.9% of mixed waste volume in Sweden (the highest percentage) and only 5.6% in Israel (the lowest). These estimates strike us as mostly reasonable, although exactly comparable estimates from other sources in volumetric terms are hard to find. For example, the US EPA estimated that, as of 2018, around 22% of municipal solid waste (MSW) in the US by weight was comprised of food waste (Krause et al., 2023). Given that food waste is the densest component of MSW, its percentage of MSW *by volume* is necessarily lower (US EPA, 2016). For instance, if we assume that food waste is twice as dense as the other components of MSW, then the 22% of food waste in MSW by weight equates to 12.4% by volume—quite close to the US estimate of 13.4% implied by our volumetric regression.¹

6 Discussion

This paper shows clear evidence across several OECD countries of a modest increase in food waste when it is easier for households to compost. Policymakers should be aware of this effect as they implement programs to divert organic waste from landfills. Many countries and cities are expanding curbside composting, often accompanied by mandates to separate food scraps. Since 2023, the European Union’s Waste Framework Directive requires member states to implement systems for the separation and composting of household ‘bio-waste’ (EU, 2025). In North America, several Canadian provinces and large cities in the US have implemented residential curbside composting

¹Denoting ρ_V and ρ_W as the fractions of food waste in MSW by volume and weight respectively, the two quantities are related by the formula $\rho_V = [1 + \alpha(\rho_W^{-1} - 1)]^{-1}$, where α is the ratio between food v. non-food waste densities.

services in conjunction with bans on landfilling organic waste. Similar efforts are being undertaken across the Middle East and Asia. Our results can inform such efforts by helping policymakers better anticipate the volume of compost that will need to be handled and processed as curbside collection expands.

At the same time, reductions in the amount of food discarded—whether or not it is separated and composted—is an independent policy goal in its own right (United Nations Environment Program, 2024). Food production systems intensively utilize land, water, and energy resources, and reducing the amount of food production needed to meet consumer demand can help decrease agriculture’s environmental footprint (Usubiaga et al., 2018; Lagerkvist et al., 2023). Our findings suggest that these two policy goals—expanding composting and decreasing food waste—may sometimes conflict. To minimize this conflict, policymakers could consider pairing curbside composting programs with additional, complementary efforts specifically aimed at reducing household food waste. For this, they may look to the behavioral intervention literature for examples (van der Werf et al., 2019; Ananda et al., 2022).

Given the observational, cross-sectional nature of the data we analyze here, there are important limitations to acknowledge in our empirical analysis. Although we took steps to mitigate selection bias in terms of which households have curbside composting, additional identification strategies should be pursued in future work. These could include better instruments, as well as the procurement and analysis of relevant panel data on household waste food waste generation pre- and post-roll-out of composting programs. In addition, a randomized controlled trial (RCT) of curbside composting implementation (*e.g.*, using staggered roll-out), in conjunction with household food waste measurements, would provide an ideal way to improve estimates of this rebound effect, study heterogeneity, and further identify the mechanisms behind it.

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Tables and figures

Table 1: Curbside composting availability and food waste index scores

		Proportion with service		Mean food waste score by service type		
<i>Country</i>	<i>N</i>	<i>No composting</i>	<i>Curbside</i>	<i>No composting</i>	<i>Curbside</i>	Δ <i>waste scores</i>
Belgium	881	0.123	0.625	0.2052	0.2299	0.0247
Canada	864	0.249	0.538	0.2202	0.2262	0.0060
France	870	0.237	0.433	0.1359	0.1861	0.0502**
Israel	866	0.507	0.085	0.4017	0.4347	0.0330
Netherlands	869	0.165	0.518	0.1807	0.2596	0.0789***
Sweden	848	0.169	0.459	0.2529	0.2618	0.0089
Switzerland	880	0.188	0.288	0.1515	0.1891	0.0376
UK	878	0.314	0.505	0.2053	0.2528	0.0475**
USA	1613	0.445	0.263	0.2738	0.3239	0.0501***

For each country, N indicates the number of survey respondents with non-missing answers to the food waste question. The * in the ‘ Δ waste score’ column indicate the statistical significance of the difference in food waste score means between the ‘No composting’ and ‘Curbside’ groups, with * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Survey questions used for environmental attitude latent class analysis

<i>To what extent do you agree with each of the following statements?</i>
Environmental impacts are frequently overstated.
I am willing to make compromises in my current lifestyle for the benefit of the environment.
Protecting the environment can boost the economy.
Environmental issues should be dealt with primarily by future generations.
Environmental issues should be resolved mainly through public policies.
Environmental policies introduced by the government should not cost me extra money.
Environmental issues will be resolved primarily through technological progress.
Environmental issues will be resolved mainly through individuals voluntarily changing their behavior.
See Brown (2024) for a description of the latent class analysis as applied to these attitudinal questions.

Table 3: Curbside composting effects (versus no composting service) on food waste index.

	(1)	(2)	(3)	(4)	(5)	(6)
Model	PSM	CPSM	CPSM	NN	OLS	Lewbel IV
Treatment effect	0.03169*** (0.01008)	0.03805*** (0.01168)	0.04087*** (0.01140)	0.02029 (0.01791)	0.03414*** (0.00797)	0.0432* (0.0226)
Country fixed effects (FE) & clustered matching	FE in PS	Within-country matching	Preferred within-country	Preferred within-country	FE	FE
Matches per obs.	2	2	2	1		
Caliper (SD)	0.15	0.15	0.15	0.2		
Dropped obs.	0	100	1	540		

Treatment effect standard errors in parentheses (* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$). Sample size = 5,838 respondents. PSM = Propensity score matching (with country fixed effects in propensity scores), CPSM = Clustered propensity score matching (Arpino and Cannas, 2016); see Section 4.1, NN = nearest neighbor matching (with Mahalanobis distance metric), OLS = ordinary least squares regression, Lewbel IV = two-stage least squares regression using heteroskedasticity-based IVs (Lewbel, 2012); see Section 4.2. Propensity score was estimated using logistic regression with the following variables: age, age², log of household size, log of house size, renter, location (rural, suburban, urban), income quantile, any kids, gender, pay-as-you-throw waste tariffs, employment type, and LCA environmental attitude probabilities (country FE for PSM model). The Mahalanobis NN matching covariates were renter, location, any kids, home composting, pay-as-you-throw waste service, log of house size, and LCA environmental attitude probabilities. Caliper (in terms of standard deviations) is the maximum distance at which two observations can be considered matches. For the propensity scores, calipers between 0.05 and 0.20 result in the same estimates. Standard errors are clustered at the country level for the OLS and Lewbel IV estimates. See the supplementary material for the underlying logistic regression results (Table S1) used to generate propensity scores, as well as complete regression estimates (Table S2) corresponding to the OLS and Lewbel results.

Table 4: Estimated effects of curbside composting (v. no composting) on volume of food discarded by households (see Section 4.3)

Food waste index transform, $f(F)$	$f(F) = F$		$f(F) = \ln(1 + \alpha F)/\alpha$	
	<i>Treatment effects</i>	<i>Food/(mixed waste)</i>	<i>Treatment effects</i>	<i>Food/(mixed waste)</i>
Belgium	3.8%	14.9%	2.0%	20.5%
Canada	2.0%	9.6%	1.3%	16.5%
France	1.9%	5.5%	0.9%	7.8%
Israel	0.3%	2.9%	0.2%	5.6%
Netherlands	3.3%	11.7%	1.6%	16.8%
Sweden	2.6%	13.5%	1.9%	27.9%
Switzerland	2.1%	7.0%	1.3%	13.1%
UK	2.4%	10.2%	1.3%	15.9%
USA	1.1%	7.0%	0.8%	13.4%
Observations	2,412		2,412	
R ²	0.375		0.384	
Log-likelihood	-2,412		-2,393	
Model parameters	45		46	
BIC	5,174		5,144	

Summary results from regressions of household mixed waste volume on food waste index value for non-composting households, using two different specifications for the transformation $f(F)$ of the food waste index F (see eq. 7). *Treatment effects* columns show the average estimated % change in volume of food discarded (as compost or mixed waste), by country (see eq. 11), combined with estimated food waste index treatment effects from col. 3 of Table 3. *Food/(mixed waste)* columns show the estimated mean % of food in total mixed waste volume, by country, for non-composting households (see eq. 9). The underlying regression estimates can be found in Table S3 of the supplementary material.

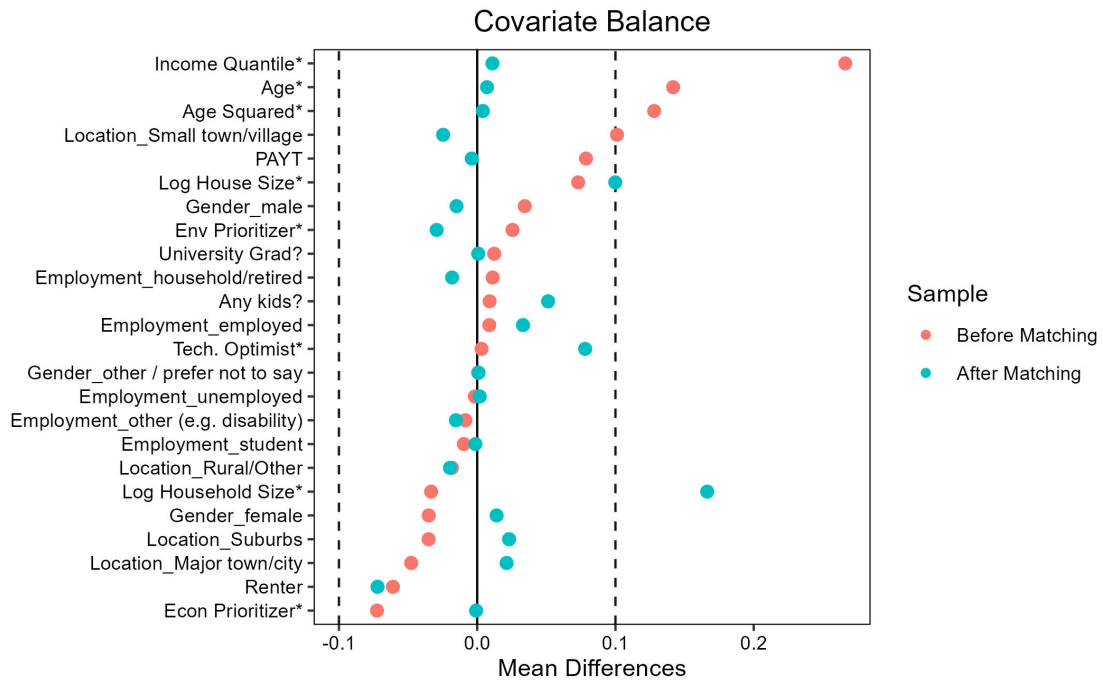


Figure 1: Covariate balance for propensity score matching with preferred within-country matching (see col. 3 of Table 3). Standardized mean differences shown.